



Gabriel Rodrigues

Bioinformatics Scientist

- ▶ Bahia, Brazil
- ▶ Full Remote
- ▶ LatAm Timezone (GMT-3)

Stack

R 5 years

Rmarkdown 5 yrs.

Python 3 yrs.

Shiny 2 yrs.

Quarto 2 yrs.

Core Skills

- ▶ NGS & Bulk RNA-Seq Analysis
- ▶ Multi-omics Integration
- ▶ Functional Genomics
- ▶ Cloud & HPC (AWS, Linux, SLURM)

Workflows

- ▶ Python (Pixi, UV)
- ▶ R (Bioconductor)
- ▶ Bash & Conda
- ▶ Nextflow Pipelines
- ▶ Machine Learning & LLMs

Professional Summary

Bioinformatics Scientist with extensive experience designing and deploying next-generation sequencing (NGS) workflows and integrating complex multi-omics datasets. Proficient in UNIX/Linux high-performance computing, Bash scripting, and orchestrating scalable data pipelines utilizing Nextflow and AWS cloud infrastructure. Demonstrated expertise in developing advanced analytical approaches—including machine learning and dimensionality reduction algorithms—to mine genomic databases for target identification. Strong track record of automating data processing and translating large-scale biological datasets into actionable insights through robust R Shiny applications.

Experience

PhD Researcher, Bioinformatics & NGS Workflows

Biotechnology and Genetics Center
UESC - Universidade Estadual de Santa Cruz

04/2025 - Present

Architected and deployed automated NGS data analysis workflows using Python, Bash, and Nextflow on AWS and Linux HPC environments (SLURM). Developed 'ViralQuest', an open-source pipeline for rapid viral sequence mining and annotation. Innovated analytical methods by implementing Isomap and machine learning algorithms for multi-omics predictive modeling.

Public Health Data Analyst & R Shiny Developer

UESC - Universidade Estadual de Santa Cruz
Regional Health Authorities

08/2025 - Present

Developed interactive data visualization tools and web applications using R Shiny to integrate and analyze large-scale disease datasets from the Notifiable Diseases Information System (SINAN). Automated reporting pipelines to deliver rigorous, actionable insights to cross-functional teams and public health stakeholders.

Genomics Researcher - Transcriptomics

Biotechnology and Genetics Center
UESC - Universidade Estadual de Santa Cruz

04/2023 - 05/2025

Performed comprehensive bulk RNA-seq data analysis to interpret functional genomic datasets. Utilized R/Bioconductor for statistical analysis, differential expression, and integration of multi-omics data. Benchmarked bioinformatics tools, ensured stringent QC measures, and contributed to peer-reviewed publications regarding novel target identification.

Key Projects and Computational Tools

Check my [GitHub profile](#) for a list of code repositories.

ViralQuest Bioinformatics Pipeline

Creator and Lead Developer

Engineered an automated, scalable Python package (published on PyPI) for end-to-end pathogen sequence identification. Integrated advanced sequence alignment and data mining methodologies across public genomic databases. Maintained rigorous technical documentation, strict version control, and modular coding standards for cross-functional scientific use. The project also utilizes D3.js resources to create a final HTML report to the final user.

Data Visualization

- ▶ R Shiny & HTML
- ▶ Quarto & RMarkdown
- ▶ Plotly, ggplot & D3.js

Education

04/2023 - 05/2025

MSc. in Genetics and Molecular Biology

Universidade Estadual de Santa Cruz

RNA-Seq Analysis of cocoa tissues

06/2018 - 08/2022

Bachelor in Interdisciplinary Sciences

Universidade Federal do Sul da Bahia

Biostatistics of Native species in Atlantic Forest

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Scalable Data Visualization Frameworks

Developer

Built dynamic R Shiny applications designed to process, analyze, and visualize high-throughput epidemiological datasets in real-time. Ensured compliance with data integrity standards while translating raw statistical outputs into accessible formats for non-technical audiences. The project uses a combination of bslib/custom CSS to styling, shinymanager for authentication and the package renv combined with docker to version control and reproducibility.

Multi-omics Data Integration & Functional Genomics

Genomic Analyst

Led end-to-end processing of transcriptomic data (processing, assembly, downstream statistical analysis) using state-of-the-art computational methods. Identified novel biological targets via comprehensive miRNA network screening and pathway characterization. Collaborated seamlessly across disciplinary boundaries to interpret complex results and co-author technical manuscripts.

Selected Publications

ViralQuest: A user-friendly interactive pipeline for viral-sequences analysis

Rodrigues, G. V. P. et al.

02/2026

BMC Bioinformatics. DOI: <https://doi.org/10.1186/s12859-026-06391-6>

Tracing ancient viral footprints: endogenous viral elements in *Bombus* species

Conceição, L. B. A., ..., Rodrigues, G. V. P. et al.

02/2026

Mobile DNA. DOI: <https://doi.org/10.1186/s13100-026-00393-0>

Insights into miRNAs of the Stingless Bee *Melipona quadrifasciata*

Soares, D. O., ..., Rodrigues, G. V. P. et al.

06/2025

Non-Coding RNA. DOI: <https://doi.org/10.3390/ncrna11030048>

Theobroma cacao Virome: Public RNA-Seq Data for Viral Discovery

Rodrigues, G. V. P. et al.

04/2025

Viruses. DOI: <https://www.mdpi.com/1999-4915/17/5/624>

The Virome of Cocoa Fermentation-Associated Microorganisms

Santos, J. P. N., Rodrigues, G. V. P. et al.

07/2024

Viruses. DOI: <https://doi.org/10.3390/v16081226>